

Learning objectives

In this lesson you will learn to:

- Describe the role of the operating system in a computer system
- Identify tasks carried out by an operating system.

For more information and additional student activities see Topic 3.2 of the student book.

What makes a computer

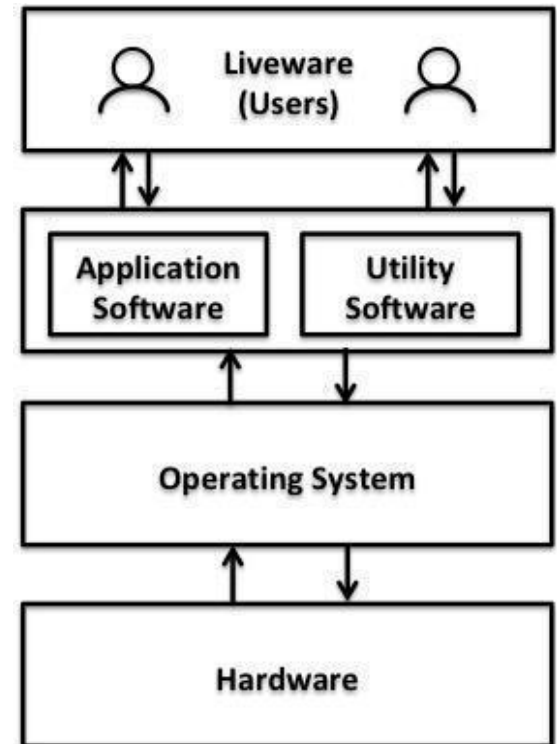
- Previously you have examined some of the hardware that makes up a computer.
- Software programs (made up of data and instructions) are stored in secondary storage, which is persistent (i.e. remains when the power is switched off), but slow to read and write.
- Programs are copied into the memory which is volatile (i.e. data is lost when the power is switched off), but has a high read and write speed.
- Then the CPU performs the fetch decode execute cycle to carry out each instruction.
- The next step is to explore the programs themselves.

Software

- There are two types of software that computers need to be useful.
- The first type is **system software**.
- These are the programs that enable a computer to function. They include the operating system and utility programs.
- The second type is **application software**.
- These are the programs that the user interacts with most. They include programs such as spreadsheets, web browsers and games.

The operating system

- So far you have looked at computer hardware and its interaction with single instructions.
- However, computer programs are much more complex than single instructions.
- The most important software in a computer system is the operating system (OS).
- Its main function is to provide a platform for other software, so that this software can interact with the computer hardware.



A wide variety

- Many types of computing device have operating systems.
- What are the names of some operating systems you know?
- The most common are:
 - Microsoft Windows (desktops, laptops and tablets)
 - macOS (Apple Macs)
 - Linux (desktops, laptops and servers).

What is an operating system?

- An operating system is a large and complex program.
- Its job is to provide an interface between the hardware and other software, and the user.
- It has a range of responsibilities to ensure the computer is usable and useful.

Operating system responsibilities

- The operating system performs many tasks.
- These fall into the following broad categories:
 - process management – organising instructions in memory so that they can be processed
 - file management – organising the contents of secondary storage
 - user management – providing a user interface and access levels
 - peripheral management – interfacing with device drivers to control physical devices.

Wrap up: you have learned how to...

- ✓ Describe the role of the operating system (OS) in a computer system.
 - The OS provides a platform on which other software can run.
 - It enables devices and software to communicate with hardware.
- ✓ Identify the tasks carried out by an OS.
 - File management.
 - User interface.
 - Process and memory management.
 - Peripheral management via device drivers.